₿ CryptoFolio — Blockchain DApp

Complete Presentation Scripts & Viva Guide | Subject: Blockchain Technology

Solidity 0.8.20 • Hardhat EVM (31337)

🎙️ 1. Manjunathreddy

PROJECT LEAD • ~1.5 MINS

🖥️ **What to show:** Open the homepage showing the title **₿ CryptoFolio** and top summary cards.

"Respected professor and fellow classmates, a very good morning/afternoon.

Today, my team—consisting of Sohail, Shivu, Sathvik, Rajesh, and myself—is presenting **CryptoFolio**, a real-time Decentralized Application (DApp) for cryptocurrency portfolio management and market tracking.

The Problem: When crypto investors hold digital assets across multiple blockchains, tracking real-time profit and loss without relying on centralized, insecure databases is difficult.

Our Solution: We developed a decentralized web application where all user investments are permanently stored on the **Ethereum Blockchain** using a **Solidity 0.8.20 Smart Contract**. It connects live market rates with on-chain transparency.

Now, I hand over to **Sohail** to walk you through our UI and blockchain architecture."

🎙️ 2. Sohail

UI & ARCHITECTURE • ~1.5 MINS

🖥️ **What to show:** Point to the top **Live Ticker Bar** and the **Blockchain System Status Banner**.

"Thank you, Manjunath.

For our interface, we used a modern, responsive **dark glassmorphic design**. At the very top, our **Live Marquee Ticker Bar** continuously scrolls current prices and 24-hour changes for 12 major cryptocurrencies in a seamless loop.

Right below the header, please look at our **Blockchain System Status Banner**:

- **EVM Node Status:** Shows '• *System Connected*' with a live indicator.
- **Smart Contract Network:** Connected to our local **Hardhat EVM (Chain ID: 31337)**.
- **Storage Architecture:** Powered by **Solidity 0.8.20** on-chain storage.
- **Connected Wallet:** Shows the active Hardhat test account `0xf39F...2266`.

Now, **Shivu** will demonstrate how we record an investment onto the blockchain."

🎙️ 3. Shivu (Shivakumar)

LIVE ON-CHAIN DEMO • ~45 SECONDS (EASY & SHORT)

🖥️ **What to show:** In the **Add Investment** form, choose *Bitcoin*, enter Qty `0.5`, Price `45000`, and click **'Add'**.

"Thank you, Sohail. Adding an asset to the blockchain is very simple:

1. I select a coin from the list—like **Bitcoin**.
2. I enter the **Quantity** and the **Buy Price** in Rupees.
3. I click the **'Add'** button.

Immediately, an EVM transaction is signed and mined into a block! The investment is saved permanently to the smart contract, and our **Holdings Table** and **Allocation Bar** update instantly.

Now, **Sathvik** will explain how we fetch live market rates."

🎙️ 4. Sathvik

MARKET DATA & SPARKLINES • ~1.5 MINS

🖥️ **What to show:** Scroll to the **Crypto Market** cards and point to the 7-day mini chart curves.

"Thank you, Shivu.

While user holdings are stored on-chain, live asset prices come from the **CoinGecko REST API**:

1. **Single Batch Request:** Instead of making 12 separate calls, we fetch prices, 24-hour volume, market cap, and 7-day historical prices in just **one consolidated call** for high speed.
2. **Auto-Refresh & Backoff:** Data refreshes every 60 seconds with an automatic retry backoff if rate-limited.
3. **Canvas Sparklines:** Look at the mini charts on each coin card: they are drawn dynamically using native **HTML5 Canvas 2D** without any heavy third-party chart libraries.

Now, **Rajesh** will explain the financial math, blockchain verification, and future scope."

🎙️ 5. Rajesh

FINANCIAL ENGINE & BLOCKCHAIN PROOF • ~1.5 MINS

🖥️ **What to show:** Point to **Total Investment & P/L** cards, and switch briefly to the **Hardhat terminal**.

"Thank you, Sathvik.

Our client automatically calculates:

- **Total Investment:** Sum of all purchase amounts.
- **Current Value:** On-chain quantities multiplied by live market prices.
- **Total Profit/Loss:** Net difference with percentage returns.

Proof of Blockchain: Notice that user data is *not* stored in browser localStorage or a centralized SQL database. Every entry lives in our **Solidity contract mapping**. You can see the transaction hashes and mined blocks live in our Hardhat terminal!

Future Scope: We plan to connect MetaMask on the Sepolia testnet and integrate decentralized price oracles like Chainlink.

That concludes our presentation. Thank you, and we are ready for any questions!"

💡 VIVA CHEAT SHEET (QUICK ANSWERS IF PROFESSOR ASKS)

Q: What blockchain network is this running on?
A: It is running on a local Ethereum node powered by Hardhat with EVM Chain ID 31337 on localhost port 8545.

Q: Which language did you use for the Smart Contract?
A: Solidity version 0.8.20. It uses structs and mappings to record user investments on-chain.

Q: How does the webpage communicate with the smart contract?
A: We use Ethers.js via JSON-RPC to sign and broadcast transactions to our Hardhat node.